

Software Engineering & Project Management Lab Experiment No: - 10

Aim: To learn Dockerfile instructions, build an image for a sample web application using DOCKERFILE.

Theory:

What is Docker?

Docker is a platform that allows developers to build, package, and deploy applications in lightweight, portable containers. These containers include everything needed to run an application, such as code, runtime, system tools, libraries, and dependencies.

Benefits of Docker

1. Portability

- Containers can run on any platform that supports Docker.
- Applications behave consistently across different environments.

2. Efficiency

- Containers share the host OS kernel, reducing overhead and improving performance.
- They consume fewer resources compared to virtual machines.

3. Isolation

- Each container runs in its own isolated environment, preventing dependency conflicts.

4. Scalability

- Applications can be scaled up quickly by launching multiple containers.
- Docker enables automatic load balancing in large-scale deployments.

5. Consistency

- Ensures that the application runs the same way in development, testing, and production.
- Eliminates the "works on my machine" problem.

DIRECTIVE argument

Although the DIRECTIVE is case-insensitive, it is recommended to write all directives in uppercase to differentiate them from arguments. A Dockerfile usually consists of multiple lines of instructions that are executed sequentially by the Docker engine during the image-building process.

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Now that you understand the purpose of a Dockerfile, let's get our hands dirty by building one for a sample Python application.

Building a Dockerfile for a Sample Python/Flask Application

The application we'll be working with is a simple Flask app with only one home route that returns Hello, World!.

Let's start by setting up the Flask application. Open your favorite code editor and create a new directory for your project. In this directory, create a new file named app.py and add the following Python code:

```
from flask import Flask
app = Flask(__name__)
@app.route('/')
def hello():
    return 'Hello, World!'
```

Now, let's build the Dockerfile.

In the same directory as your app.py, create a new file named Dockerfile (with no file extension). This is where you'll write the instructions for Docker to build your image.

Now, follow the steps below to create the Dockerfile:

#1: Specify the base image

The very first instruction you write in a Dockerfile must be the FROM directive, which specifies the base image. The base image is the image from which all other layers in your Docker image will be built. It's the foundation of your Docker image, much like the foundation of a building.

Add the following instruction to your Dockerfile:

```
FROM python:3.11-slim
```

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Here, we're telling Docker to use the official Python Docker image, and more specifically, the 3.11-slim version. This slim variant of Python Docker image is a minimal version that excludes some packages to make the image smaller. Note that the base image you specify will be downloaded from Docker Hub, Docker's official image registry. The reason we're using Python as the base image is that the application containerization is written in Python. The Python base image includes a Python interpreter, which is necessary to run Python code, as well as a number of commonly used Python utilities and libraries. By using the Python base image, we're ensuring that the environment within the Docker container is preconfigured for running Python code.

#2 Set the working directory

Once you've chosen the base image, the next step is to determine the working directory using the WORKDIR directive. Insert the following line after the FROM directive:

```
WORKDIR /app
```

Here, we're telling Docker to create a directory named app in the Docker container and use it as the current working directory. All instructions that FROM python:3.11-slim WORKDIR /app follow (like RUN, COPY, and CMD) will be run in this directory inside the container. Think of this as typing the command cd /app in a terminal to change the current working directory to /app. The difference here is that it's being done within the Docker container as part of the build process. A working directory within the container is necessary because it designates a specific location for our application code within the container and determines where commands will be run from. If we don't set a working directory, Docker won't have a clear context for where your application is located, which would make it harder to interact with.

#3 Install dependencies Once the working directory is set, the next step is to install the dependencies. Our Python application relies on the Flask web framework, which manages requests, routes URLs, and handles other web-related tasks. To install Flask, add the following instruction in your Dockerfile just under the WORKDIR directive:

```
RUN pip install flask==2.3
```

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Here, we're instructing Docker to use pip (a package installer for Python) to install the specific version of Flask we need for our application.

#4 Copy application files to the container After setting up the working directory and installing the necessary dependencies, we're now ready to copy the application files into the Docker container. To do this, add the following instruction just below the RUN directive:

```
COPY . /app
```

This line copies everything in the current directory (denoted by ".") on our host machine into the /app directory we previously set as our working directory within the Docker container. It's like using the cp command in the terminal to copy files from one directory to another, but in this context, it's copying files from your local machine to the Docker container. Why do we need to do this? It's simple. Without this step, the Docker container wouldn't have access to our application's code, making it impossible to run our app.

#5 Specify the environment variable Once the application files are copied, we need to set up the FLASK_APP environment variable for our Docker container using the ENV directive. Now, you may be wondering why we need this environment variable in the first place. In our app.py file, we create an instance of the Flask application and assign it to the variable app. This application instance is what Flask needs to run, and it's located in the app.py file. When starting our Flask application using the flask run command (which we'll discuss in the next section), Flask must know where to locate the application instance to run. Flask uses the FLASK_APP environment variable to find this instance. Hence, we need to use the ENV directive to set the value of FLASK_APP to app.py. To do this, add the following line under the RUN directive:

```
ENV FLASK_APP=app.py
```

This line ensures Flask knows exactly where to find the application instance to run, which in our case is app.py.

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#6 Define the default command The last instruction that we need for our application is to specify the default command that will be executed when the Docker container starts:

```
CMD ["flask", "run", "--host=0.0.0.0", "--port=5000"]
```

from the image, we'll build from this Dockerfile.

Insert the following instruction below the ENV directive:

Here's what each part of the argument passed to the CMD directive does:

- flask: This is the program that we want to run. In this case, it's the Flask command-line interface.
- run: This command instructs Flask to start a local development server.
- --host=0.0.0.0: This argument tells the Flask server to listen on all public IPs. In the context of Docker, this means the Flask application will be accessible on any IP address that can reach the Docker container.
- --port=5000: This argument specifies the port number that the Flask server will listen on. Port 5000 is the default port for Flask, but it's good practice to explicitly declare it for clarity.

After this, our Dockerfile is ready. It should look like this:

```
FROM python:3.11-slim
WORKDIR /app
RUN pip install flask==2.3
COPY . /app
ENV FLASK_APP=app.py
CMD ["flask", "run", "--host=0.0.0.0", "--port=5000"]
```

It's worth noting that the directives we used in the Dockerfile for our Python app aren't the only ones available in Docker. But they are the ones you'll often encounter when working with Dockerfiles.

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#7 Create a .dockerignore file Before we go ahead and build our Docker image, we need to take care of one last thing. Remember the following COPY directive?

```
COPY ./app
```

This line instructs Docker to copy everything from our current directory to the app directory inside the container, which includes the Dockerfile itself. But, the Dockerfile isn't required for our app to work—it's just for us to create the Docker image. So, we need to ensure that the Dockerfile doesn't get copied to the app directory in the container. Here's how we do it:

Create a new file called .dockerignore in the same directory as your Dockerfile. This file works much like a .gitignore file if you're familiar with Git. Then, add the word Dockerfile to this file. This tells Docker to ignore the Dockerfile when copying files into the container. Now that we've prepared everything, it's time to build our Docker image, run a container from this image, and test our application to see if everything works as expected.

Building and running the Docker Image

Open a terminal and navigate to the directory where your Dockerfile is located. Now, run the following command to create an image named sample-flask-app:v1 (you can name the image anything you prefer):

```
$ docker build . -t sample-flask-app:v1
```

In the command above, the dot (.) after the build command indicates that the current directory is the build context. We're using the -t flag to tag the Docker image with the name sample-flask-app and version v1. After running this command, you'll see an output similar to this:

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```
$ docker build . -t sample-flask-app:v1
[+] Building 17.7s (10/10) FINISHED docker:default
=> [internal] load build definition from Dockerfile 0.0s
=> => transferring dockerfile: 192B 0.0s
=> [internal] load metadata for docker.io/library/python:3.11 3.5s
=> [auth] library/python:pull token for registry-1.docker.io 0.0s
=> [internal] load .dockerignore 0.0s
=> => transferring context: 50B 0.0s
=> [1/4] FROM docker.io/library/python:3.11 9.0s
=> => resolve docker.io/library/python:3.11 0.0s
=> => sha256:1103112ebfc46e 3.51MB / 3.51MB 1.9s
=> => sha256:b4b80ef7128d 12.87MB / 12.87MB 2.7s
=> => sha256:a2eb07f336e4f1 1.65kB / 1.65kB 0.0s
=> => sha256:4bcdb5d5bc81ca 1.37kB / 1.37kB 0.0s
=> => sha256:15646a3fa12dde 6.93kB / 6.93kB 0.0s
=> => sha256:8a1e25ce7c4f 29.12MB / 29.12MB 4.8s
=> => sha256:cc7f04ac52f8a3bad5 243B / 243B 2.4s
=> => sha256:87b8bf94a2ace2 3.41MB / 3.41MB 3.3s
=> => extracting sha256:8a1e25ce7c4f75e372e 1.8s
=> => extracting sha256:1103112ebfc46e01c0f 0.2s
=> => extracting sha256:b4b80ef7128dc9bd114 1.0s
=> => extracting sha256:cc7f04ac52f8a3bad5b 0.0s
=> => extracting sha256:87b8bf94a2ace2b005d 0.7s
=> [internal] load build context 0.0s
=> => transferring context: 194B 0.0s
=> [2/4] WORKDIR /app 0.2s
=> [3/4] RUN pip install flask==2.3 4.7s
=> [4/4] COPY . /app 0.0s
=> exporting to image 0.2s
=> => exporting layers 0.2s
=> => writing image sha256:c6879156c7750c89 0.0s
=> => naming to docker.io/library/sample-fl 0.0s

What's Next?
View a summary of image vulnerabilities and recommendations → docker scout quickview
```

To make sure the image sample-flask-app:v1 has been successfully created, run the following command to check the list of Docker images on your system:

```
$ docker image ls
```


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The resulting output should look something like this:

```
$ docker image ls
```

REPOSITORY	TAG	IMAGE ID	CREATED	SIZE
sample-flask-app	v1	c6879156c775	10 seconds ago	147MB
mongo	latest	24041ceefc56	6 days ago	755MB

In the list, you should see sample-flask-app:v1, which confirms the image is now in our system.

Now, run the sample-flask-app:v1 image as a container by executing the following command:

```
$ docker container run -d -p 5000:5000 sample-flask-app:v1
```

The -d flag is short for --detach and runs the container in the background. The -p flag is short for --publish and maps port 5000 of the host to port 5000 of the Docker container. After running this command, you'll see an output like this:

```
$ docker container run -d -p 5000:5000 sample-flask-app:v1  
ff37071dd4cef95cc1dc2ce7e145019339cfaec54575659f72aea4e560238f8c
```

The long string you see printed in the terminal is the container ID. To make sure the container is running, list the currently active Docker containers by running the following command:

```
$ docker container ls
```

You should see something like this:

```
$ docker container ls
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
c301c50152ac	sample-flask-app:v1	"flask run --host=0.0.0.0"	14 seconds ago	Up 12 seconds	0.0.0.0:5000->5000/tcp	xenodochial_mendel

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The container is up and running as expected. Our Flask application is now running inside the container. To test it, open a web browser and go to <http://localhost:5000>. You should see the message Hello, World! displayed like this:



SCREENSHOTS:

```
ubuntu@ip-172-31-40-218: ~  
ubuntu@ip-172-31-40-218:~$  
ubuntu@ip-172-31-40-218:~$ systemctl status docker  
● docker.service - Docker Application Container Engine  
   Loaded: loaded (/lib/systemd/system/docker.service; enabled; vendor preset: enabled)  
   Active: active (running) since Mon 2023-04-10 19:46:05 UTC; 18min ago  
 TriggeredBy: ● docker.socket  
     Docs: https://docs.docker.com  
    Main PID: 684 (dockerd)  
       Tasks: 12  
      Memory: 141.2M  
         CPU: 3.736s  
    CGroup: /system.slice/docker.service  
            └─684 /usr/bin/dockerd -H fd:// --containerd=/run/containerd/containerd.sock  
  
Apr 10 19:46:01 ip-172-31-40-218 dockerd[684]: time="2023-04-10T19:46:01.899794323Z" level=info msg="ccResolverWrapper: se  
Apr 10 19:46:01 ip-172-31-40-218 dockerd[684]: time="2023-04-10T19:46:01.899956895Z" level=info msg="ClientConn switching  
Apr 10 19:46:02 ip-172-31-40-218 dockerd[684]: time="2023-04-10T19:46:02.339992511Z" level=info msg="[graphdriver] using p  
Apr 10 19:46:03 ip-172-31-40-218 dockerd[684]: time="2023-04-10T19:46:03.665306795Z" level=info msg="Loading containers: s  
Apr 10 19:46:04 ip-172-31-40-218 dockerd[684]: time="2023-04-10T19:46:04.873139021Z" level=info msg="Default bridge (docke  
Apr 10 19:46:05 ip-172-31-40-218 dockerd[684]: time="2023-04-10T19:46:05.081644328Z" level=info msg="Loading containers: d  
Apr 10 19:46:05 ip-172-31-40-218 dockerd[684]: time="2023-04-10T19:46:05.543882435Z" level=info msg="Docker daemon" commit  
Apr 10 19:46:05 ip-172-31-40-218 dockerd[684]: time="2023-04-10T19:46:05.547797680Z" level=info msg="Daemon has completed  
Apr 10 19:46:05 ip-172-31-40-218 systemd[1]: Started Docker Application Container Engine.  
Apr 10 19:46:05 ip-172-31-40-218 dockerd[684]: time="2023-04-10T19:46:05.743749833Z" level=info msg="API listen on /run/do  
lines 1-22/22 (END)  
[  
ubuntu@ip-172-31-40-218:~$ docker ps  
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS   NAMES  
ubuntu@ip-172-31-40-218:~$ docker images  
REPOSITORY    TAG       IMAGE ID   CREATED   SIZE  
ubuntu@ip-172-31-40-218:~$
```

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```
ubuntu@ip-172-31-40-218:~$  
ubuntu@ip-172-31-40-218:~$ pwd  
/home/ubuntu  
ubuntu@ip-172-31-40-218:~$ mkdir my-website  
ubuntu@ip-172-31-40-218:~$ cd my-website/  
ubuntu@ip-172-31-40-218:~/my-website$ wget https://www.free-css.com/assets/files/free-css-templates/download/page290/wave-cafe.zip  
--2023-04-10 20:06:14-- https://www.free-css.com/assets/files/free-css-templates/download/page290/wave-cafe.zip  
Resolving www.free-css.com (www.free-css.com)... 217.160.0.242, 2001:8d8:100f:f000::28f  
Connecting to www.free-css.com (www.free-css.com)|217.160.0.242|:443... connected.  
HTTP request sent, awaiting response... 200 OK  
Length: 11896390 (11M) [application/zip]  
Saving to: 'wave-cafe.zip'  
  
wave-cafe.zip          100%[=====>] 11.34M  6.08MB/s   in 1.9s  
  
2023-04-10 20:06:17 (6.08 MB/s) - 'wave-cafe.zip' saved [11896390/11896390]  
  
ubuntu@ip-172-31-40-218:~/my-website$  
ubuntu@ip-172-31-40-218:~/my-website$  
ubuntu@ip-172-31-40-218:~/my-website$ ls  
wave-cafe.zip  
ubuntu@ip-172-31-40-218:~/my-website$ unzip wave-cafe.zip
```

```
ubuntu@ip-172-31-40-218: ~/my-website  
inflating: 2121_wave_cafe/fontawesome/webfonts/fa-regular-400.ttf  
inflating: 2121_wave_cafe/fontawesome/webfonts/fa-regular-400.woff  
inflating: 2121_wave_cafe/fontawesome/webfonts/fa-regular-400.woff2  
inflating: 2121_wave_cafe/fontawesome/webfonts/fa-solid-900.eot  
inflating: 2121_wave_cafe/fontawesome/webfonts/fa-solid-900.svg  
inflating: 2121_wave_cafe/fontawesome/webfonts/fa-solid-900.ttf  
inflating: 2121_wave_cafe/fontawesome/webfonts/fa-solid-900.woff  
inflating: 2121_wave_cafe/fontawesome/webfonts/fa-solid-900.woff2  
creating: 2121_wave_cafe/img/  
inflating: 2121_wave_cafe/img/about-1.png  
inflating: 2121_wave_cafe/img/about-2.png  
inflating: 2121_wave_cafe/img/hot-americano.png  
inflating: 2121_wave_cafe/img/hot-cappuccino.png  
inflating: 2121_wave_cafe/img/hot-espresso.png  
inflating: 2121_wave_cafe/img/hot-latte.png  
inflating: 2121_wave_cafe/img/iced-americano.png  
inflating: 2121_wave_cafe/img/iced-cappuccino.png  
inflating: 2121_wave_cafe/img/iced-espresso.png  
inflating: 2121_wave_cafe/img/iced-latte.png  
inflating: 2121_wave_cafe/img/smoothie-1.png  
inflating: 2121_wave_cafe/img/smoothie-2.png  
inflating: 2121_wave_cafe/img/smoothie-3.png  
inflating: 2121_wave_cafe/img/smoothie-4.png  
inflating: 2121_wave_cafe/img/special-01.jpg  
inflating: 2121_wave_cafe/img/special-02.jpg  
inflating: 2121_wave_cafe/img/special-03.jpg  
inflating: 2121_wave_cafe/img/special-04.jpg  
inflating: 2121_wave_cafe/img/special-05.jpg  
inflating: 2121_wave_cafe/img/special-06.jpg  
inflating: 2121_wave_cafe/index.html  
creating: 2121_wave_cafe/js/  
inflating: 2121_wave_cafe/js/jquery-3.4.1.min.js  
creating: 2121_wave_cafe/video/  
inflating: 2121_wave_cafe/video/wave-cafe-video-bg.mp4  
ubuntu@ip-172-31-40-218:~/my-website$  
ubuntu@ip-172-31-40-218:~/my-website$  
ubuntu@ip-172-31-40-218:~/my-website$  
ubuntu@ip-172-31-40-218:~/my-website$ clear
```

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```
ubuntu@ip-172-31-40-218: ~/my-website
ubuntu@ip-172-31-40-218:~/my-website$
ubuntu@ip-172-31-40-218:~/my-website$ ls
2121_wave_cafe  wave-cafe.zip
ubuntu@ip-172-31-40-218:~/my-website$ cd 2121_wave_cafe
ubuntu@ip-172-31-40-218:~/my-website/2121_wave_cafe$ ls
css  fontawesome  img  index.html  js  video
ubuntu@ip-172-31-40-218:~/my-website/2121_wave_cafe$ cp -R * ../
ubuntu@ip-172-31-40-218:~/my-website/2121_wave_cafe$
ubuntu@ip-172-31-40-218:~/my-website/2121_wave_cafe$
ubuntu@ip-172-31-40-218:~/my-website/2121_wave_cafe$ cd ..
ubuntu@ip-172-31-40-218:~/my-website$ ls
2121_wave_cafe  css  fontawesome  img  index.html  js  video  wave-cafe.zip
ubuntu@ip-172-31-40-218:~/my-website$ rm -rf wave-cafe.zip 2121_wave_cafe
ubuntu@ip-172-31-40-218:~/my-website$
ubuntu@ip-172-31-40-218:~/my-website$ ls
css  fontawesome  img  index.html  js  video
ubuntu@ip-172-31-40-218:~/my-website$ nano Dockerfile
```

```
GNU nano 6.2 Dockerfile
FROM httpd:2.4
COPY . /usr/local/apache2/htdocs/

[ Wrote 2 lines ]
^G Help      ^O Write Out  ^W Where Is   ^K Cut        ^T Execute    ^C Location   M-U Undo     M-A Set Mark
^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justify    ^_ Go To Line  M-E Redo     M-6 Copy
```

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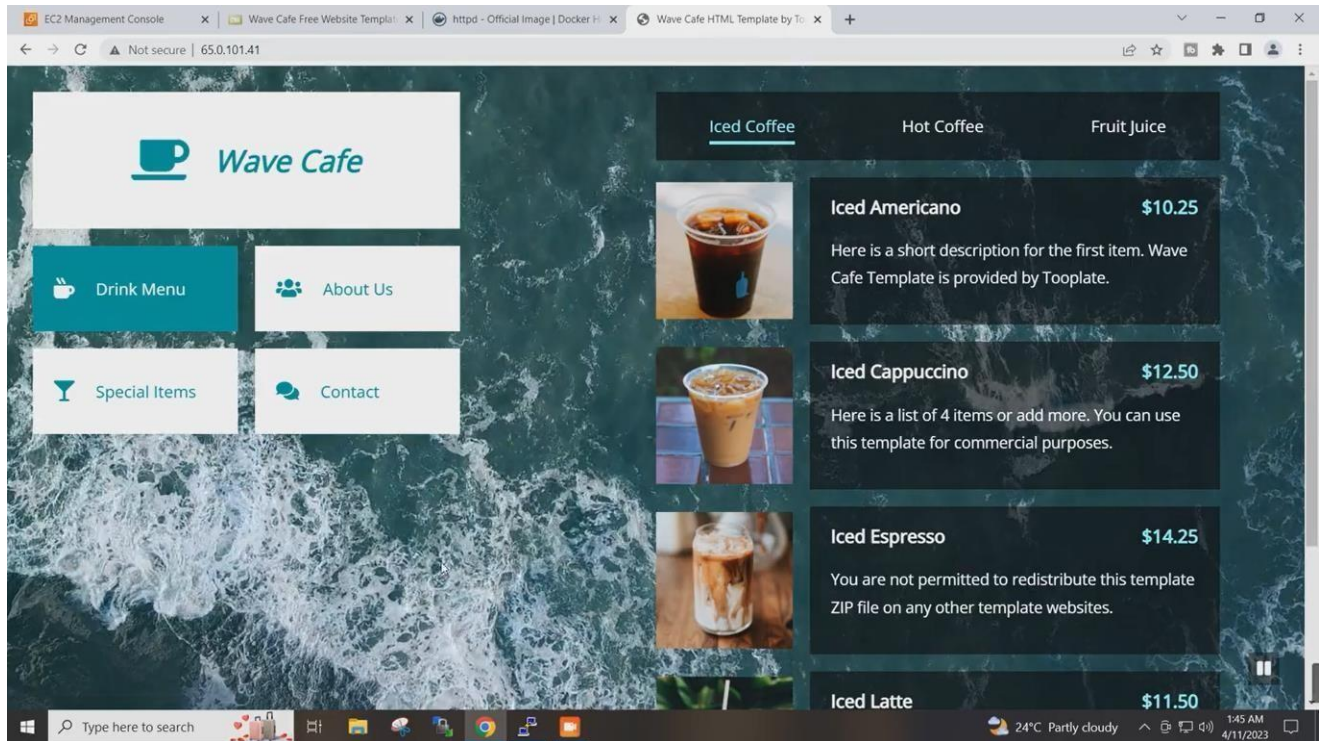
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```
ubuntu@ip-172-31-40-218:~/my-website$
ubuntu@ip-172-31-40-218:~/my-website$ ls
2121_wave_cafe  wave-cafe.zip
ubuntu@ip-172-31-40-218:~/my-website$ cd 2121_wave_cafe
ubuntu@ip-172-31-40-218:~/my-website/2121_wave_cafe$ ls
css fontawesome img index.html js video
ubuntu@ip-172-31-40-218:~/my-website/2121_wave_cafe$ cp -R * ../
ubuntu@ip-172-31-40-218:~/my-website/2121_wave_cafe$
ubuntu@ip-172-31-40-218:~/my-website/2121_wave_cafe$ cd ..
ubuntu@ip-172-31-40-218:~/my-website$ ls
2121_wave_cafe css fontawesome img index.html js video wave-cafe.zip
ubuntu@ip-172-31-40-218:~/my-website$ rm -rf wave-cafe.zip 2121_wave_cafe
ubuntu@ip-172-31-40-218:~/my-website$
ubuntu@ip-172-31-40-218:~/my-website$ ls
css fontawesome img index.html js video
ubuntu@ip-172-31-40-218:~/my-website$ nano Dockerfile
Dockerfile css fontawesome img index.html js video
ubuntu@ip-172-31-40-218:~/my-website$ docker build . -t my-website:latest
Sending build context to Docker daemon 13.61MB
Step 1/2 : FROM httpd:2.4
2.4: Pulling from library/httpd
f1f26f570256: Pull complete
a6b093ae1967: Pull complete
6b400bbb27df: Pull complete
6e310dd059b6: Pull complete
471cb5914961: Pull complete
Digest: sha256:4055b18d92fd006f74d4a2aac172a371dc9a750eaa78000756dee55a9beeb4625
Status: Downloaded newer image for httpd:2.4
--> dcl95e13784
Step 2/2 : COPY . /usr/local/apache2/htdocs/
--> 7d48427f5e2f
Successfully built 7d48427f5e2f
Successfully tagged my-website:latest
ubuntu@ip-172-31-40-218:~/my-website$
ubuntu@ip-172-31-40-218:~/my-website$
ubuntu@ip-172-31-40-218:~/my-website$ clear
```

```
ubuntu@ip-172-31-40-218: ~/my-website
ubuntu@ip-172-31-40-218:~/my-website$
ubuntu@ip-172-31-40-218:~/my-website$ docker images
REPOSITORY TAG IMAGE ID CREATED SIZE
my-website latest 7d48427f5e2f 15 seconds ago 159MB
httpd 2.4 dcl95e13784 4 days ago 145MB
ubuntu@ip-172-31-40-218:~/my-website$
ubuntu@ip-172-31-40-218:~/my-website$ docker run -d -p 80:80 my-website:latest
e0a6d7f3ab6718a1b648d9b5f00dcc89e846d1fe12bd568ce9b1412fc0d3c9da
ubuntu@ip-172-31-40-218:~/my-website$
ubuntu@ip-172-31-40-218:~/my-website$
ubuntu@ip-172-31-40-218:~/my-website$ docker ps
CONTAINER ID IMAGE COMMAND CREATED STATUS PORTS
e0a6d7f3ab67 my-website:latest "httpd-foreground" 8 seconds ago Up 7 seconds 0.0.0.0:80->80/tcp, :::80->80/tcp
trusting_rosalind
ubuntu@ip-172-31-40-218:~/my-website$
```


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Conclusion: Thus, we have successfully learnt Dockerfile instructions & build an image for a sample web application using DOCKERFILE.